HAOYANG YU

LOW-DIMENSIONAL TOPOLOGY · FLOER HOMOLOGY

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School of Mathematical Sciences, Peking University

EDUCATION

Peking University 2023 – 2027 (expected)

B.S. in Mathematics and Applied Mathematics · Beijing, China

RESEARCH INTERESTS

Low-dimensional topology; knot theory; 3-manifold topology; Heegaard Floer homology; monopole Floer homology.

RESEARCH EXPERIENCE

The University of Chicago Mathematics REU Jun – Aug 2026

Participant · Chicago, Illinois

- Completed an eight-week program of intensive mathematical study and research under the mentorship of Daniel Minahan, culminating in a final paper.

3-Manifolds and Floer Homology Sep 2025 – Nov 2026

Undergraduate Research Project · School of Mathematical Sciences, Peking University

- Studied Morse homology, Heegaard Floer homology, monopole Floer homology and 3-manifold topology under the supervision of Prof. Jiajun Wang.

SELECTED PROJECTS

Interactive Knot-Theory Visualizations 2026

Independent browser-based software projects

- **Knot Isotopy Explorer.** A visualization of PL knots that constrains vertex motion with Δ-move intersection tests, redraws planar diagrams in real time, detects Reidemeister moves, and computes the Alexander polynomial from a Wirtinger presentation via Fox calculus.
- **Lifting a Knot from Its Diagram.** A tool that reconstructs an embedded spatial PL knot from a planar shadow and over/under data with exact reprojection; for at most 12 crossings it enumerates all crossing assignments and groups them by Alexander polynomial.

HONORS AND AWARDS

2026	National Scholarship, Peking University (2025–2026 academic year)
2026	Ubiquant Mathematics Talent Scholarship, Peking University (2025–2026 academic year)
2026	8th Place, Geometry and Topology Written Examination, 17th S.-T. Yau College Student Mathematics Contest
2025, 2026	Merit Student, Peking University (2024–2025 and 2025–2026 academic years)
2025	Honorable Mention, Written Examination in Geometry and Topology, 16th S.-T. Yau College Student Mathematics Contest
2025	First Prize, 36th Beijing College Students' Mathematics Competition (Mathematics Major, Group A)
2025	Qin Wanshun–Jin Yunhui Scholarship, Peking University (2024–2025 academic year)
2025	Young Talent Support Program in Basic Mathematics, School of Mathematical Sciences, Peking University (supported by Xiwang)
2021, 2022	Gold Medal (First Prize), Chinese Mathematical Olympiad (CMO)